

IND320-OLEGR099\D2Dbook\streamlit app\Pages\First_month.py

```
1 import streamlit as st
2 import pandas as pd
3
4 @st.cache_data
5 def load_reservoir_data():
6     data = pd.read_csv("../data/reservoirs_processed.csv")
7
8     data = data.rename(columns={
9         "dato_Id": "Date",
10        "fyllingsgrad": "filling_level",
11        "kapasitet_TWh": "Capacity(TWh)",
12        "fylling_TWh": "Filled (TWh)",
13        "fyllingsgrad_forrige_uke": "filling_level_previous_week",
14        "endring_fyllingsgrad": "change_filling_level"
15    })
16
17    data["Date"] = pd.to_datetime(data["Date"])
18
19    return data
20
21
22 reservoir_data = load_reservoir_data()
23
24 st.title("Reservoirs")
25
26 # First month
27 first_month = reservoir_data[
28     reservoir_data["Date"] < reservoir_data["Date"].min() + pd.DateOffset(months=1)
29 ]
30
31 chart_data = pd.DataFrame({
32     "Variable": [
33         "filling_level",
34         "Capacity(TWh)",
35         "Filled (TWh)",
36         "filling_level_previous_week",
37         "change_filling_level"
38     ],
39     "Values": [
40         first_month["filling_level"].tolist(),
41         first_month["Capacity(TWh)"].tolist(),
42         first_month["Filled (TWh)"].tolist(),
43         first_month["filling_level_previous_week"].tolist(),
44         first_month["change_filling_level"].tolist()
45     ]
46 })
47
48 st.dataframe(
49     chart_data,
50     column_config={
51         "Values": st.column_config.LineChartColumn(
```

```
52 |         "First month",
53 |         width="large"
54 |     )
55 | },
56 | hide_index=True
57 | )
58 |
59 |
```